Customer Spend Prediction & Platform Optimization

Predicting Annual Customer Spending Using Machine Learning

Author: Fisayo Olawale Ajayi

Tools: Python | Pandas | NumPy | Matplotlib | Seaborn | Scikit-learn

Project Type: Regression Analysis | Predictive Analytics | Customer Analytics

Executive Summary

Understanding the factors that influence customer spending is essential for improving customer retention, increasing customer lifetime value, and optimizing marketing investments.

This project develops a regression model to predict annual customer spending using customer engagement metrics collected from an e-commerce platform. Beyond prediction, the analysis identifies which customer behaviors contribute most to long-term revenue, providing actionable insights for business decision-making.

The project follows a structured analytics workflow that includes exploratory data analysis, feature evaluation, regression modeling, model validation, and interpretation of business findings.

Business Problem

Customer acquisition is expensive, making customer retention and lifetime value key drivers of sustainable business growth.

Although businesses collect large volumes of customer engagement data, it is often unclear which behaviors translate into higher spending.

By identifying the strongest predictors of annual spending, organizations can focus their resources on initiatives that increase customer value while improving marketing efficiency.

Business Objectives

The objectives of this project are to:

- Explore customer engagement behavior.
- Identify factors influencing annual customer spending.
- Build a regression model capable of predicting yearly spending.
- Evaluate model performance using appropriate regression metrics.
- Translate analytical findings into business recommendations.

Analytical Questions

This analysis seeks to answer the following questions:

1. Which customer characteristics most influence annual spending?
2. Can yearly customer spending be predicted accurately using behavioral data?
3. Which engagement channel contributes most to customer value?
4. How can these findings support customer retention and revenue optimization?

Dataset Overview

The dataset contains customer demographic information and engagement metrics collected across multiple digital channels.

Each record represents an individual customer and includes information related to account tenure, website usage, mobile application usage, session duration, and annual spending.

The target variable is **Yearly Amount Spent**, while the remaining variables serve as predictors in the regression model.

Double-click (or enter) to edit

Importing the libraries

Environment Setup

This section imports the libraries required for data manipulation, visualization, statistical analysis, and machine learning. These tools support the end-to-end analytical workflow, from data preparation through predictive modeling and evaluation.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import sklearn as sc


%matplotlib inline
```

```
Ecommerce_Customers = pd.read_csv('https://raw.githubusercontent.com/Fisayo-Ajayi/fisayo/main/Ecommerce%20Customers')
```

Data Quality Assessment

Before developing predictive models, the dataset is examined for missing values, data types, and overall quality.

This assessment ensures the dataset is suitable for analysis and identifies any preprocessing steps required before model development.

```
Ecommerce_Customers.describe()
```

	Avg. Session Length	Time on App	Time on Website	Length of Membership	Yearly Amount Spent
count	500.000000	500.000000	500.000000	500.000000	500.000000
mean	33.053194	12.052488	37.060445	3.533462	499.314038
std	0.992563	0.994216	1.010489	0.999278	79.314782
min	29.532429	8.508152	33.913847	0.269901	256.670582
25%	32.341822	11.388153	36.349257	2.930450	445.038277
50%	33.082008	11.983231	37.069367	3.533975	498.887875
75%	33.711985	12.753850	37.716432	4.126502	549.313828
max	36.139662	15.126994	40.005182	6.922689	765.518462

```
Ecommerce_Customers.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 8 columns):
 #   Column                Non-Null Count  Dtype  
---  ------                --------------  -----  
 0   Email                 500 non-null    object 
 1   Address               500 non-null    object 
 2   Avatar                500 non-null    object 
 3   Avg. Session Length   500 non-null    float64
 4   Time on App           500 non-null    float64
 5   Time on Website       500 non-null    float64
 6   Length of Membership  500 non-null    float64
 7   Yearly Amount Spent   500 non-null    float64
dtypes: float64(5), object(3)
memory usage: 31.4+ KB
```

EXPLORATORY DATA ANALYSIS

Observation

State what the chart shows.

Business Interpretation

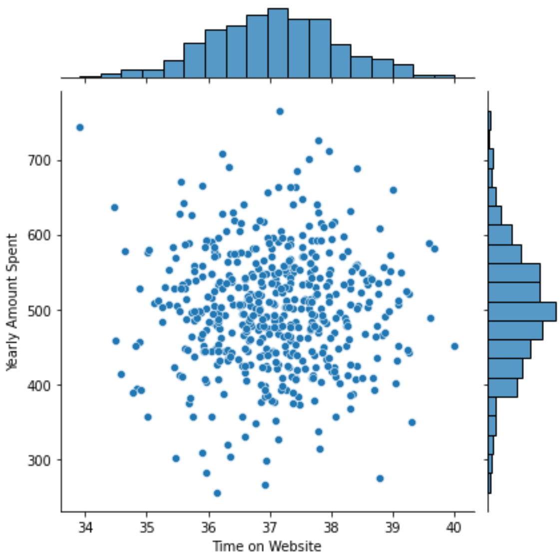
Explain why it matters.

Potential Business Impact

Describe how the insight could influence customer retention, marketing, pricing, or another business decision.

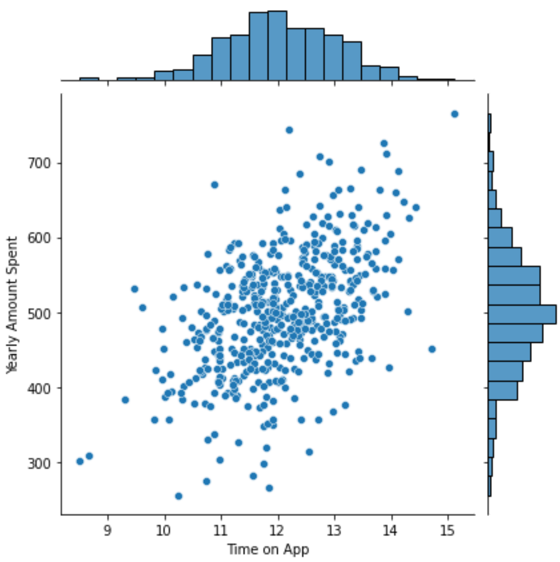
```
sns.jointplot(data=Ecommerce_Customers, x='Time on Website' ,y='Yearly Amount Spent' )
```

```
<seaborn.axisgrid.JointGrid at 0x7f8de6920650>
```




```
sns.jointplot(data=Ecommerce_Customers, x= 'Time on App', y ='Yearly Amount Spent')
```

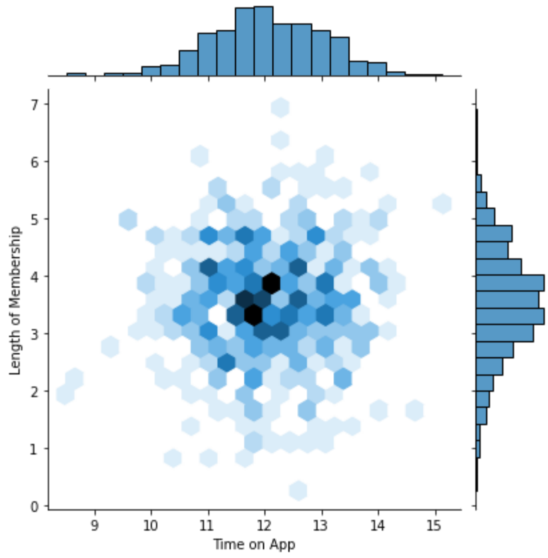
```
<seaborn.axisgrid.JointGrid at 0x7f8de3a04ed0>
```



** Use jointplot to create a 2D hex bin plot comparing Time on App and Length of Membership.**

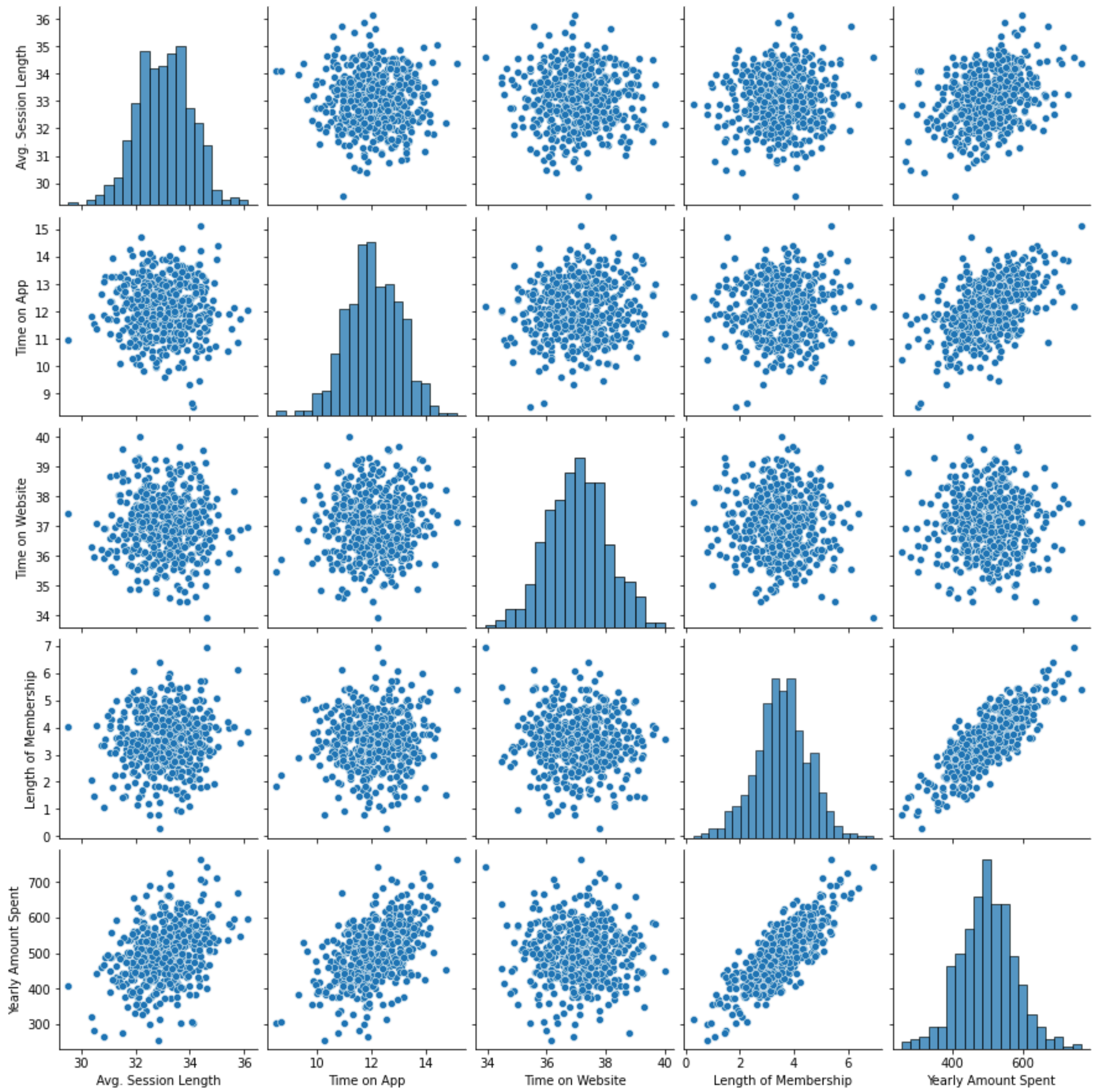
```
sns.jointplot(data=Ecommerce_Customers , x= 'Time on App', y = 'Length of Membership', kind = 'hex')
```

```
<seaborn.axisgrid.JointGrid at 0x7f8de3888890>
```




```
sns.pairplot(Ecommerce_Customers)
```

```
<seaborn.axisgrid.PairGrid at 0x7f8de372cd90>
```

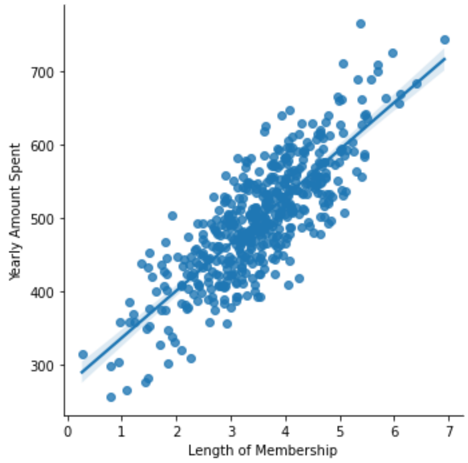


Based off this plot what looks to be the most correlated feature with Yearly Amount Spent?

```
# Length of Membership
```

```
sns.lmplot(data=Ecommerce_Customers, x = 'Length of Membership' , y="Yearly Amount Spent")
```

```
<seaborn.axisgrid.FacetGrid at 0x7f8de2ad8650>
```



Training and Testing Data

Now that we've explored the data a bit, let's go ahead and split the data into training and testing sets. ** Set a variable X equal to the numerical features of the customers and a variable y equal to the "Yearly Amount Spent" column. **

```
Ecommerce_Customers.columns
```

```
Index(['Email', 'Address', 'Avatar', 'Avg. Session Length', 'Time on App',
       'Time on Website', 'Length of Membership', 'Yearly Amount Spent'],
      dtype='object')
```

```
y = Ecommerce_Customers['Yearly Amount Spent']
```

```
X = Ecommerce_Customers[[ 'Avg. Session Length', 'Time on App',
       'Time on Website', 'Length of Membership']]
```

** Use model_selection.train_test_split from sklearn to split the data into training and testing sets. Set test_size=0.3 and random_state=101**

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=101)
```

Model Development

A regression model is developed to estimate annual customer spending using behavioral and engagement features. The objective is to understand how customer characteristics relate to spending while producing reliable predictions that support business planning.

```
from sklearn.linear_model import LinearRegression
```

Double-click (or enter) to edit

```
lm = LinearRegression()
```

```
lm.fit(X_train, y_train)
print(lm.intercept_)
```

```
-1047.9327822502391
```

Print out the coefficients of the model

```
lm.coef_
```

```
array([25.98154972, 38.59015875,  0.19040528, 61.27909654])
```

Predicting Test Data

Now that we have fit our model, let's evaluate its performance by predicting off the test values!

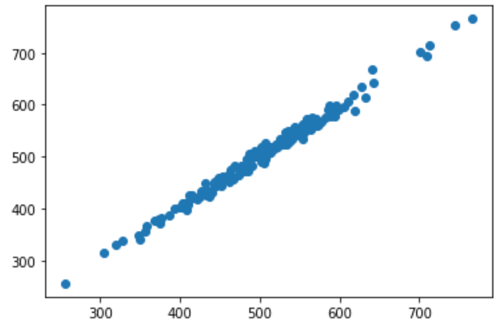
** Use lm.predict() to predict off the X_test set of the data.**

```
Predictions = lm.predict(X_test)
```

** Create a scatterplot of the real test values versus the predicted values. **

```
plt.scatter(y_test, Predictions)
```

```
<matplotlib.collections.PathCollection at 0x7f8de2afd590>
```



Evaluating the Model

Let's evaluate our model performance by calculating the residual sum of squares and the explained variance score (R^2).

** Calculate the Mean Absolute Error, Mean Squared Error, and the Root Mean Squared Error. **

```
from sklearn import metrics
```

```
print('MAE' , metrics.mean_absolute_error(y_test, Predictions))
print('MSE', metrics.mean_squared_error(y_test, Predictions))
print('RMSE', np.sqrt(metrics.mean_squared_error(y_test, Predictions)))
```

```
MAE 7.228148653430826
MSE 79.81305165097427
RMSE 8.933815066978624
```

```
metrics.explained_variance_score(y_test, Predictions)
```

```
0.9890771231889606
```

Model Performance

The model achieved strong predictive performance, indicating that the selected features explain a substantial proportion of the variation in customer spending. Performance is assessed using R^2, MAE, and RMSE, providing complementary perspectives on model accuracy and prediction error.

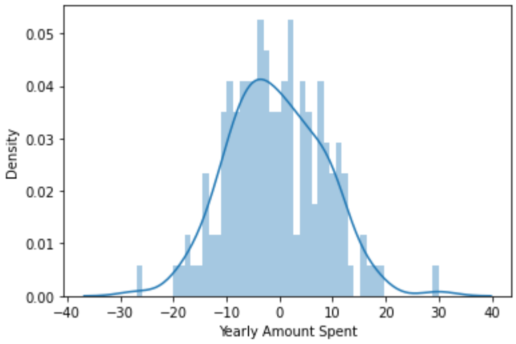
Residual Analysis

Model evaluation metrics provide a quantitative assessment of prediction accuracy. Residual analysis complements these metrics by examining the distribution of prediction errors.

Ideally, residuals should be randomly distributed around zero and approximately follow a normal distribution. This indicates that the regression model is capturing the underlying relationships without introducing systematic bias.

```
sns.distplot((y_test-Predictions), bins=50)
```

```
/usr/local/lib/python3.7/dist-packages/seaborn/distributions.py:2619: FutureWarning: `distplot` is a deprecated function and
  warnings.warn(msg, FutureWarning)
<matplotlib.axes._subplots.AxesSubplot at 0x7f8ddddfcc50>
```




```
CDF= pd.DataFrame(lm.coef_, X.columns, columns=['coeff'])
CDF.head()
```

	coeff
Avg. Session Length	25.981550
Time on App	38.590159
Time on Website	0.190405
Length of Membership	61.279097

** How can we interpret these coefficients? **

From the table values above, a unit increase in on the time on the app gives a coefficient of 38 plus units in price, and this means we hve more monetary inputs coming from the influx of time spent on the app, i would recommend the companies and make more developments that would further increase members time spent and in turn increase profits